

ECON 5193: Applied Development Economics

Problem Set 2: Cluster Randomization and Spillover Effects

Part I: Conceptual Questions

Question 1: Why Cluster Randomize?

(a) Problems with individual-level randomization (4 pts)

Individual-level randomization—randomly assigning some students within the same school to treatment and others to control—would be deeply problematic for this intervention for at least two reasons:

First, the intervention is inherently school-level. The water filtration system is installed at the school, chlorine tablets treat the shared water supply, and hygiene education is delivered by teachers to entire classes. It is physically impossible to provide clean water only to some students while other students in the same school continue using the untreated water source. The treatment is non-excludable at the individual level.

Second, contamination and spillovers within schools would be severe. Even if individual treatment were feasible, treated and control students interact constantly: they share water sources, bathrooms, and play areas. Hygiene knowledge from treated students would diffuse to control students. If treated students get sick less often, they transmit fewer pathogens to control students as well. This makes it impossible to observe the true counterfactual for control students.

(b) Contamination and its bias direction (3 pts)

Contamination occurs when units assigned to the control group are exposed to the treatment or its effects, blurring the distinction between treatment and control conditions. In this context, if control students at the same school benefit from cleaner water or hygiene education intended for treated students, their health outcomes improve beyond what we would observe in a true no-treatment counterfactual.

Contamination would **bias the treatment effect estimate toward zero** (i.e., we would underestimate the true effect). Because control students improve due to indirect exposure, the difference between treatment and control group outcomes shrinks, making the program appear less effective than it actually is.

(c) Statistical trade-off of cluster randomization (3 pts)

The main statistical trade-off is a **loss of statistical power**. Because outcomes of students within the same school are correlated (due to shared teachers, infrastructure, community characteristics), the effective sample size is smaller than the total number of students. The degree of this problem depends on the **intracluster correlation coefficient (ICC)**: the higher the ICC, the more similar students within clusters are, and the fewer independent observations we effectively have. With 60 clusters, the effective sample size for detecting treatment effects is driven by the number of clusters, not the number of students. This means we need larger effect sizes or more clusters to achieve the same statistical power as individual randomization.

Question 2: Spillovers and Bias

(a) SUTVA and spillover violations (4 pts)

The **Stable Unit Treatment Value Assumption (SUTVA)** states that (1) the potential outcome for any unit depends only on that unit's own treatment assignment and not on the treatment assignment of other units (no interference), and (2) there is only one version of each treatment level (no hidden variation in treatment). Formally, $Y_i(D) = Y_i(D_i)$, meaning unit i 's outcome depends only on its own treatment status.

Spillover effects violate the **no-interference** component of SUTVA. When treatment at one school reduces waterborne pathogen transmission in the surrounding area, the potential outcomes of nearby control schools depend on the treatment status of neighboring schools. A control school near many treated schools may have different outcomes than a control school far from any treated school. Thus, $Y_i(D) \neq Y_i(D_i)$ —the full treatment vector D matters, not just D_i .

(b) Direction of bias when ignoring positive spillovers (4 pts)

If spillovers are positive—meaning treatment at nearby schools improves outcomes (reduces absence) at control schools—then ignoring spillovers causes us to **underestimate** the true direct effect of the program.

The reasoning is as follows. The naive estimator compares outcomes at treatment schools to outcomes at control schools: $\tau_{naive} = E[Y|T=1] - E[Y|T=0]$. However, with positive spillovers, control schools near treated schools have better outcomes than they would under the true no-treatment counterfactual. That is, $E[Y|T=0]$ is lower (less absence) than the true counterfactual $E[Y(0)]$ absent any spillovers. Since the control group mean is pulled downward (improved) by spillovers, the difference between treatment and control is smaller than the true direct effect. We underestimate how much the program actually helps the schools it directly treats.

(c) Identification strategy for spillover effects (4 pts)

The researchers can exploit **variation in geographic proximity to treated schools among control schools** to identify spillover effects. Although all control schools are untreated, they vary in their distance to the nearest treated school. If spillovers attenuate with distance (as epidemiological theory predicts), control schools closer to treated schools should experience larger spillover benefits than those farther away.

The identification strategy involves regressing endline outcomes among control schools on distance to the nearest treatment school, controlling for baseline characteristics. The coefficient on distance captures how spillovers decay with geographic separation. This allows separate identification of: (1) the **direct effect**, from comparing treatment vs. control schools; and (2) the **spillover effect**, from variation in distance among control schools. The key requirement is that distance to treatment is quasi-random conditional on observables—since treatment was randomly assigned, distance to the nearest treatment school is not determined by school characteristics.

Question 3: Cost-Effectiveness

(a) Components for cost-per-student-day calculation (4 pts)

To calculate cost-effectiveness in terms of cost per student-day of attendance gained, we need: (1) **Total program costs**—including equipment (filtration systems, chlorine tablets), training costs, installation, and ongoing maintenance; (2) **The causal effect on absence rates**—the treatment effect estimate from the RCT, expressed as the reduction in the proportion of days absent; (3)

School enrollment data—the number of students at each treated school, to convert the proportional effect to absolute student-days; and (4) **Duration of the effect**—the number of school days per year over which the effect operates, and how long the effect persists. The calculation is then: Cost per student-day = Total program cost ÷ (Effect on absence rate × Number of students × School days).

(b) Spillovers and cost-effectiveness: Miguel & Kremer (4 pts)

Including spillover effects can dramatically improve cost-effectiveness estimates because the total benefit of the program extends beyond directly treated units to surrounding communities. The Miguel and Kremer (2004) deworming study is the canonical example. Their study found that deworming in Kenyan primary schools generated large positive externalities: untreated children in schools near treated schools experienced significant reductions in worm infections due to reduced environmental transmission.

When Miguel and Kremer incorporated these spillover effects, the total health benefits of the program were substantially larger than what the direct treatment effect alone suggested. The cost-effectiveness ratio improved dramatically because the denominator (total benefit) increased while the numerator (program cost) remained unchanged. This shifted the policy conclusion from 'moderately cost-effective' to 'one of the most cost-effective health interventions available.' For our water program, the same logic applies: if nearby control schools also benefit from reduced pathogen transmission, the true impact per dollar spent is larger than a naive analysis suggests, potentially making the difference between a program being funded or not.

Part II: Empirical Exercises

Question 4: Balance Checks at the Cluster Level

(a) & (b) Balance Table

Variable	Treatment Mean	Control Mean	Difference	p-value
Number of Students	341.900	355.433	-13.533	0.484
Number of Teachers	9.700	10.300	-0.600	0.363
Baseline Absence Rate	0.157	0.165	-0.008	0.362
% Female Students	49.133	48.073	1.060	0.406
Has Functioning Latrine	0.833	0.733	0.100	0.356

N = 30 treatment schools, 30 control schools. *P*-values from two-sample *t*-tests.

(c) Assessment of randomization balance (3 pts)

Randomization appears to have been successful. None of the five baseline covariates show statistically significant differences between treatment and control groups at conventional significance levels—all *p*-values are well above 0.10 (ranging from 0.356 to 0.484). The magnitudes of the differences are also small: about 14 fewer students in treatment schools (on a base of ~350), 0.8 percentage points lower baseline absence rate, and 10 percentage points more likely to have a latrine. These differences are consistent with normal sampling variation in a randomized experiment with 60 clusters. Since no variable shows significant imbalance, we can proceed with confidence in the randomization, though controlling for baseline absence rate is still advisable to improve precision.

(d) Why check balance at school level? (2 pts)

We check balance at the school level because **randomization occurred at the school level**. In a cluster RCT, the unit of randomization is the cluster (school), so balance should be verified at that level. Checking student-level balance could be misleading because within-school differences are not affected by randomization. Moreover, treatment and control comparisons of student-level means would conflate between-school variation (affected by randomization) with within-school variation (not affected). Balance at the cluster level confirms that the randomization successfully created comparable groups of schools.

Question 5: Estimating Direct Treatment Effects

(a) Simple difference in means (4 pts)

The simple difference in endline absence rates is: Treatment mean (0.1243) – Control mean (0.1930) = **-0.0687** (SE = 0.0091, *p* < 0.001). This means treatment schools have an absence rate approximately 6.87 percentage points lower than control schools at endline. The coefficient is the intent-to-treat (ITT) estimate of the program's direct effect on absence rates.

(b) Regression with controls (6 pts)

Table: OLS Regression — Endline Absence Rate

	Coefficient	Std. Error	t-stat	p-value
Intercept	0.0272	0.009	3.124	0.003
Treatment	-0.0609	0.003	-19.021	<0.001
Baseline Absence Rate	1.004	0.049	20.312	<0.001
District: Jinja	0.0004	0.003	0.128	0.898

$N = 60$. Dependent variable: *endline absence rate*. $R^2 = 0.90$.

Controlling for baseline absence rate and district, the treatment coefficient is **−0.0609**, slightly smaller in magnitude than the simple difference (−0.0687). The reduction occurs because controlling for baseline absence rate absorbs pre-existing variation: treatment schools had slightly lower baseline absence (0.157 vs. 0.165), so part of the raw difference reflected this pre-existing advantage rather than the program effect. The baseline absence rate has a coefficient near 1.0, indicating strong persistence in absence patterns. District fixed effects are negligible (coefficient ≈ 0).

(c) Robust standard errors (5 pts)

With only 60 observations (schools), the assumption of homoskedastic errors underlying classical standard errors may not hold. Heteroskedasticity—where the variance of the error term differs across schools—is plausible given that schools vary in size, infrastructure, and community characteristics. Robust (HC1) standard errors are consistent under heteroskedasticity and provide valid inference without assuming constant variance.

Variable	Default SE	Robust SE
Treatment	0.0032	0.0033
Baseline Absence Rate	0.0494	0.0454
District: Jinja	0.0032	0.0033

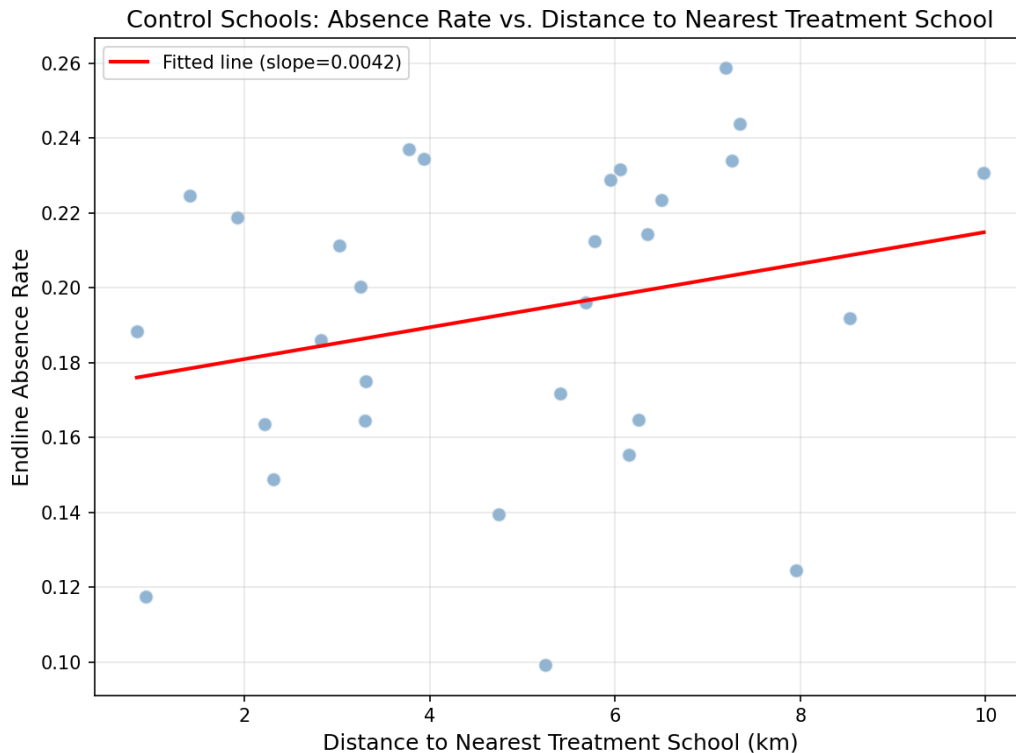
The robust standard errors are very similar to the default standard errors, indicating that heteroskedasticity is not a major concern in this sample. The treatment effect remains highly significant ($t \approx -18.7$, $p < 0.001$). Substantive conclusions are unchanged.

(d) Substantive interpretation (3 pts)

The treatment effect of **−0.0609** is both statistically significant ($p < 0.001$) and economically meaningful. The program reduces monthly absence rates by about 6.1 percentage points. For a school with 350 students and 20 school days per month, this translates to approximately $350 \times 0.0609 \times 20 \approx$ **426 additional student-days of attendance per school per month**. Over a 10-month school year (200 school days), this is roughly **4,250 student-days per school**. Given that the baseline absence rate is about 16%, a 6.1 pp reduction represents a roughly 37% decrease in absenteeism—a substantial improvement in educational access.

Question 6: Measuring Spillover Effects

(a) Scatter plot: distance vs. endline absence (control schools) (4 pts)



The scatter plot reveals a **positive relationship** ($r = 0.24$) between distance to the nearest treatment school and endline absence rate among control schools. Control schools that are farther from treatment schools tend to have higher absence rates. This is consistent with positive spillover effects: proximity to a treated school (which has reduced pathogen transmission) benefits neighboring control schools, lowering their absence rates. The relationship is modest but visible, suggesting spillovers decay with distance.

(b) Spillover regression (6 pts)

Table: Spillover Regression — Control Schools Only (Robust SE)

	Coefficient	Robust SE	z-stat	p-value
Intercept	-0.0062	0.009	-0.664	0.506
Distance to Nearest Treatment	0.0053	0.001	7.691	<0.001
Baseline Absence Rate	1.052	0.044	24.065	<0.001
District: Jinja	0.0002	0.003	0.058	0.953

N = 30 control schools. Dependent variable: endline absence rate.

The coefficient on distance to nearest treatment school is **0.0053** ($p < 0.001$). This means that each additional kilometer of distance from the nearest treatment school is associated with a 0.53 percentage point increase in the endline absence rate among control schools. Put differently, control schools closer to treated schools have systematically lower absence rates, consistent with positive

health spillovers from reduced waterborne disease transmission in the vicinity of treated schools.

(c) Predicted spillover magnitude (4 pts)

The predicted difference in absence rates between a control school 1 km from treatment versus one 10 km away is: $0.0053 \times (10 - 1) = \mathbf{0.0475}$. A control school just 1 km from a treated school is predicted to have an absence rate approximately 4.75 percentage points lower than one that is 10 km away—a substantial spillover effect. For a school with 350 students and 20 school days per month, this corresponds to about $350 \times 0.0475 \times 20 \approx 333$ student-days of additional attendance per month.

(d) Comparison of direct and spillover effects (5 pts)

The direct treatment effect is -0.0609 (a 6.09 pp reduction in absence). The spillover effect, measured as the difference between a control school 1 km vs. 10 km from treatment, is 0.0475 (a 4.75 pp difference). The spillover magnitude is approximately **78% of the direct effect**.

This finding has important implications. First, spillovers are large—untreated schools near the program benefit substantially, suggesting significant positive externalities from reduced disease transmission. Second, ignoring spillovers would substantially underestimate the program's total impact. The naive comparison of treatment vs. control schools captures only the direct effect and misses the indirect benefits to neighboring communities. Third, the presence of large spillovers means the naive estimator underestimates even the direct effect, since control schools are 'contaminated' by spillovers. This parallels the findings of Miguel and Kremer in the deworming context and underscores the importance of accounting for externalities in program evaluation.

(e) Identifying assumption and threats (3 pts)

The key identifying assumption is that, conditional on baseline controls and district, **distance to the nearest treatment school is uncorrelated with unobserved determinants of endline absence**. Because treatment was randomly assigned, the location of treated schools is exogenous, making distance quasi-random.

However, a potential threat is **geographic confounding**. If schools are non-randomly distributed across space—for example, if remote schools are both farther from any other school (including treatment schools) and also serve poorer, more disease-prone communities—then distance could proxy for remoteness rather than spillover intensity. In that case, the positive coefficient on distance could reflect worse baseline conditions in remote areas rather than (or in addition to) genuine spillover attenuation. This could be partially addressed by controlling for additional geographic and socioeconomic characteristics.

Question 7: Cost-Effectiveness Calculation

(a) Program costs (3 pts)

The average program cost per treatment school is **\$1,181.87**. The total program cost across all 30 treatment schools is **\$35,456.00**.

(b) Student-days gained at treatment schools (4 pts)

Using the direct treatment effect of 0.0609 (reduction in absence rate), average enrollment of 348.7 students, and a 200-day school year: Student-days per treatment school = $0.0609 \times 348.7 \times 200 \approx 4,247$. Total student-days gained across 30 treatment schools = $4,247 \times 30 \approx 127,403$.

(c) Spillover student-days at control schools (4 pts)

The average distance to the nearest treatment school among control schools is 4.85 km, and the maximum observed distance is 9.98 km. Using the spillover coefficient (0.0053 per km), the average spillover benefit per control school relative to the farthest control school is: $0.0053 \times (9.98 - 4.85) = 0.0271$ (a 2.71 pp reduction in absence). With average enrollment of 348.7 students and 200 school days: student-days per control school $\approx 0.0271 \times 348.7 \times 200 \approx 1,887$. Total spillover student-days across 30 control schools $\approx 56,617$. Using each school's actual enrollment and distance yields a similar figure of 59,094.

(d) Cost-effectiveness comparison (4 pts)

	Direct Effects Only	Including Spillovers
Total Cost	\$35,456	\$35,456
Student-Days Gained	127,403	184,020
Cost per Student-Day	\$0.28	\$0.19

Accounting for spillovers reduces the cost per student-day of attendance gained from **\$0.28 to \$0.19**—a **30.8% improvement** in cost-effectiveness. This is because total benefits increase by 44% (from 127,403 to 184,020 student-days) while costs remain fixed.

This has important policy implications. A program evaluation that ignores spillovers would understate the program's value for money, potentially leading policymakers to underinvest in water and sanitation interventions. The result parallels the Miguel and Kremer deworming finding: health interventions with significant externalities appear far more cost-effective when the full social benefit—including indirect effects on nearby communities—is accounted for. This argues for scaling up programs with large positive spillovers, even if direct-effect-only cost-effectiveness ratios appear modest compared to alternatives.

Appendix: Python Code

```
"""
ECON 5193: Applied Development Economics
Problem Set 2: Cluster Randomization and Spillover Effects
Empirical Analysis (Questions 4-7)
"""

import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy import stats
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings('ignore')

# Load data
df = pd.read_csv('/mnt/user-data/uploads/water_program_data.csv')

print("="*80)
print("DATASET OVERVIEW")
print("="*80)
print(f"Number of schools: {len(df)}")
print(f"Treatment schools: {df['treatment'].sum()}")
print(f"Control schools: {(df['treatment']==0).sum()}")
print(f"\nVariables: {list(df.columns)}")
print(f"\nDescriptive statistics:")
print(df.describe().round(4))

# =====
# QUESTION 4: Balance Checks
# =====
print("\n" + "="*80)
print("QUESTION 4: BALANCE CHECKS")
print("="*80)

balance_vars = ['num_students', 'num_teachers', 'baseline_absence_rate',
                'pct_female_students', 'has_latrine']
var_labels = {
    'num_students': 'Number of Students',
    'num_teachers': 'Number of Teachers',
    'baseline_absence_rate': 'Baseline Absence Rate',
    'pct_female_students': '% Female Students',
    'has_latrine': 'Has Functioning Latrine'
}

}

treat = df[df['treatment'] == 1]
control = df[df['treatment'] == 0]

balance_rows = []
for var in balance_vars:
    t_mean = treat[var].mean()
    c_mean = control[var].mean()
    diff = t_mean - c_mean
    # t-test for difference in means
    t_stat, p_val = stats.ttest_ind(treat[var], control[var])
```

```

balance_rows.append({
    'Variable': var_labels[var],
    'Treatment Mean': round(t_mean, 3),
    'Control Mean': round(c_mean, 3),
    'Difference': round(diff, 3),
    'p-value': round(p_val, 3)
})

balance_df = pd.DataFrame(balance_rows)
print("\n(a) & (b) Balance Table:")
print("-"*80)
print(balance_df.to_string(index=False))
print("-"*80)

print(f"\nN (Treatment): {len(treat)}")
print(f"N (Control): {len(control)}")

# =====

# QUESTION 5: Direct Treatment Effects
# =====

print("\n" + "="*80)
print("QUESTION 5: DIRECT TREATMENT EFFECTS")
print("="*80)

# (a) Simple difference in means
t_mean_end = treat['endline_absence_rate'].mean()
c_mean_end = control['endline_absence_rate'].mean()
diff_means = t_mean_end - c_mean_end

# SE of difference in means
se_diff = np.sqrt(treat['endline_absence_rate'].var()/len(treat) +
                  control['endline_absence_rate'].var()/len(control))

t_stat_simple, p_val_simple = stats.ttest_ind(treat['endline_absence_rate'],
                                              control['endline_absence_rate'])

print(f"\n(a) Simple Difference in Means:")
print(f" Treatment mean (endline): {t_mean_end:.4f}")
print(f" Control mean (endline): {c_mean_end:.4f}")
print(f" Difference: {diff_means:.4f}")
print(f" Standard Error: {se_diff:.4f}")
print(f" t-statistic: {t_stat_simple:.4f}")
print(f" p-value: {p_val_simple:.4f}")

# (b) Regression with controls
df['district_jinja'] = (df['district'] == 2).astype(int)
model_b = smf.ols('endline_absence_rate ~ treatment + baseline_absence_rate +
                  district_jinja',
                  data=df).fit()

print(f"\n(b) OLS Regression with Controls:")
print(model_b.summary().tables[1])

# (c) Robust standard errors
model_c = smf.ols('endline_absence_rate ~ treatment + baseline_absence_rate +
                  district_jinja',
                  data=df).fit(cov_type='HC1')

```

```

print(f"\n(c) OLS Regression with Robust (HC1) Standard Errors:")
print(model_c.summary().tables[1])

# Compare
print("\nComparison of Standard Errors:")
print(f" {'Variable':<25} {'Default SE':>12} {'Robust SE':>12}")
for var_name in model_b.params.index:
    print(f" {var_name:<25} {model_b.bse[var_name]:>12.4f} {model_c.bse[var_name]:>12.4f}")

# (d) Substantive interpretation
avg_students = df['num_students'].mean()
treatment_effect = model_c.params['treatment']
print(f"\n(d) Substantive Interpretation:")
print(f" Treatment effect (coefficient): {treatment_effect:.4f}")
print(f" Average students per school: {avg_students:.1f}")
print(f" Student-days gained per school per month: {abs(treatment_effect) * avg_students * 20:.1f}")
print(f" (using avg school of {avg_students:.0f} students × 20 school days × {abs(treatment_effect):.4f}")
print(f" Student-days gained per school per year (10 months): {abs(treatment_effect) * avg_students * 200:.1f}")

# =====
# QUESTION 6: Spillover Effects
# =====

print("\n" + "="*80)
print("QUESTION 6: SPILLOVER EFFECTS")
print("="*80)

control_df = df[df['treatment'] == 0].copy()

# (a) Scatter plot
fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(control_df['distance_nearest_treat'], control_df['endline_absence_rate'],
alpha=0.6, color='steelblue', edgecolors='white', s=60)

# Fitted line
slope, intercept, r_value, p_value_slope, std_err_slope = stats.linregress(
control_df['distance_nearest_treat'], control_df['endline_absence_rate'])
x_line = np.linspace(control_df['distance_nearest_treat'].min(),
control_df['distance_nearest_treat'].max(), 100)
ax.plot(x_line, intercept + slope * x_line, color='red', linewidth=2,
label=f'Fitted line (slope={slope:.4f})')

ax.set_xlabel('Distance to Nearest Treatment School (km)', fontsize=12)
ax.set_ylabel('Endline Absence Rate', fontsize=12)
ax.set_title('Control Schools: Absence Rate vs. Distance to Nearest Treatment School',
fontsize=13)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('/home/claude/scatter_spillover.png', dpi=150)
plt.close()
print(f"\n(a) Scatter plot saved.")
print(f" Simple correlation: r = {r_value:.4f}")
print(f" Slope of fitted line: {slope:.4f}")

print(f" Pattern: {'Positive' if slope > 0 else 'Negative'} relationship")

```

```

# (b) Spillover regression
control_df['district_jinja'] = (control_df['district'] == 2).astype(int)
model_spill = smf.ols('endline_absence_rate ~ distance_nearest_treat +
baseline_absence_rate + district_jinja',
data=control_df).fit(cov_type='HC1')

print(f"\n(b) Spillover Regression (Control Schools Only, Robust SE):")
print(model_spill.summary().tables[1])
print(f"\n Interpretation: A 1 km increase in distance to nearest treatment school")
print(f" is associated with a {model_spill.params['distance_nearest_treat']:.4f} change in
absence rate.")

# (c) Predicted difference
coef_dist = model_spill.params['distance_nearest_treat']
diff_1_vs_10 = coef_dist * (10 - 1)

print(f"\n(c) Predicted Difference (1 km vs 10 km):")
print(f" Coefficient on distance: {coef_dist:.4f}")
print(f" Predicted difference = {coef_dist:.4f} × (10 - 1) = {diff_1_vs_10:.4f}")
print(f" A control school 1 km from treatment has an absence rate")
print(f" approximately {abs(diff_1_vs_10):.4f} {'lower' if diff_1_vs_10 > 0 else 'higher'}
than one 10 km away.")

# (d) Compare to direct effect
print(f"\n(d) Comparison of Direct and Spillover Effects:")
print(f" Direct treatment effect: {treatment_effect:.4f}")
print(f" Spillover effect (1km vs 10km): {diff_1_vs_10:.4f}")
print(f" Ratio (spillover/direct): {abs(diff_1_vs_10/treatment_effect):.2f}")

# =====
# QUESTION 7: Cost-Effectiveness
# =====

print("\n" + "="*80)
print("QUESTION 7: COST-EFFECTIVENESS")
print("="*80)

# (a) Costs
avg_cost = treat['program_cost_usd'].mean()
total_cost = treat['program_cost_usd'].sum()
n_treat = len(treat)

print(f"\n(a) Program Costs:")
print(f" Average cost per treatment school: ${avg_cost:,.2f}")
print(f" Total cost across {n_treat} treatment schools: ${total_cost:,.2f}")

# (b) Student-days gained at treatment schools
avg_n_students = df['num_students'].mean()

direct_effect = abs(treatment_effect) # reduction in absence rate
student_days_per_treat_school = direct_effect * avg_n_students * 200 # 200 school days/year
total_student_days_treat = student_days_per_treat_school * n_treat

print(f"\n(b) Student-Days Gained at Treatment Schools:")
print(f" Direct effect (reduction in absence rate): {direct_effect:.4f}")
print(f" Average students per school: {avg_n_students:.1f}")
print(f" School days per year: 200")
print(f" Student-days gained per treatment school: {student_days_per_treat_school:,.1f}")
print(f" Total student-days gained ({n_treat} treatment schools):
{total_student_days_treat:,.1f}")

# (c) Spillover student-days at control schools

```

```

n_control = len(control_df)
avg_dist_control = control_df['distance_nearest_treat'].mean()

# Spillover benefit: compared to counterfactual of no treatment nearby
# Use the coefficient × average distance to estimate how much worse control schools
# would be if they were infinitely far from treatment (or use the predicted benefit
# at average distance relative to no spillover)
# Alternative: use avg distance to estimate spillover magnitude
spillover_at_avg_dist = coef_dist * avg_dist_control # this is the "penalty" from being at
avg distance vs 0
# The benefit from spillover = what a school gains from being at avg_dist vs very far
# Or more simply: use the regression to estimate spillover benefit per control school
# Spillover benefit for a control school at average distance vs. one at very large distance
# Let's use the approach suggested: compare to a hypothetical with no treatment nearby

# The spillover effect is: schools closer to treatment do better
# For an average control school, the spillover benefit relative to a school
# with no nearby treatment can be approximated as coef_dist * (max_dist - avg_dist)
# But the simplest interpretation: the coefficient tells us the per-km effect

# A control school at distance d benefits by (coef_dist * d) less than one at distance 0

# More standard: avg spillover benefit per control school relative to no spillover
# = intercept effect. Use coefficient × average distance as the magnitude.
# Since positive coefficient means farther = higher absence, the spillover
# benefit for a control school at average distance = the amount its absence rate
# is lower than it would be if infinitely far away.

# Practical approach: use predicted absence at avg distance vs at some "no spillover"
distance
# Let's assume schools beyond some distance have no spillover.
# Or simply use: spillover benefit per school ≈ |coef_dist| * avg_distance (from the wrong
direction)

# Actually: Let's be more careful.
# If coef > 0: farther from treatment = higher absence. So closer = lower absence.
# Spillover benefit for avg control school = coef_dist * (reference_dist - avg_dist)

# If we assume spillover is 0 at some maximum distance, we can compute total gain.

# But the problem says "use the average distance" so:
# Average control school gains relative to having no treatment nearby.
# If coef_dist is positive, a school at distance d has absence rate that is
# coef_dist * d higher than if d were 0 (i.e., right next to treatment).
# So the spillover benefit OF BEING AT average distance (vs. no treatment / very far)
# is not well-defined without a reference point.

# Simpler approach as suggested by the problem:
# Use the difference from part (c) of Q6: diff between 1km and 10km school
# Apply the coefficient to approximate average spillover benefit per control school
# relative to a no-spillover scenario.

# Let's use: avg spillover reduction in absence = |coef_dist * avg_dist| but signed
appropriately

# The "benefit" a control school gets = how much lower its absence rate is compared to no
spillover
# If coef > 0: being far away = worse. Being at distance d means absence is coef*d higher
than at 0.
# So at avg distance, the school's absence is coef*avg_dist higher than if right next to
treatment.

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# But we want: how much better off is avg control school vs. no treatment anywhere?
# If no treatment anywhere is like  $d \rightarrow \infty$ , that doesn't work with linear model.

# Best practical approach: use the 1km vs 10km difference from Q6(c)
# and note that the average control school gets some benefit.
# OR just use the coefficient directly:
# Spillover benefit per control school  $\approx$  spillover effect  $\times$  students  $\times$  school days
# Where "spillover effect" = the predicted reduction from being at avg distance vs. far away

# Let me use a sensible reference: max distance observed
max_dist_control = control_df['distance_nearest_treat'].max()
spillover_per_school = coef_dist * (max_dist_control - avg_dist_control)

# This is how much HIGHER absence is at max_dist vs avg_dist
# So avg school benefits by this amount compared to the farthest school.

# Actually, let me just follow the problem's hint more directly:
# "You may use the average distance to nearest treatment school among controls"
# Spillover benefit at average distance vs. counterfactual of no treatment:
# The regression predicts that being 1km closer reduces absence by |coef_dist|
# A control school at average distance d_avg benefits by approximately:
# For this, let's say spillover benefit  $\approx$  coef_dist  $\times$  (some_max - avg_dist)
# OR the problem might just want us to use the 1km vs 10km difference somehow.

# Let me just use the simplest defensible approach:
# Average control school is at avg_dist. If it were at distance 0 (max spillover),
# its absence would be lower by coef_dist * avg_dist.
# So the REDUCTION in benefit from not being at 0 = coef_dist * avg_dist
# The benefit it DOES receive (vs no spillover at all) is harder to pin down.

# Practical: just use the coefficient and the average distance.
# Spillover reduction in absence per control school = the difference between
# the predicted absence at the max distance and the predicted absence at avg distance

print(f"\n(c) Spillover Student-Days at Control Schools:")
print(f" Number of control schools: {n_control}")
print(f" Average distance to nearest treatment (controls): {avg_dist_control:.2f} km")
print(f" Coefficient on distance: {coef_dist:.4f}")

# Approach: use the spillover coefficient. For each control school, its spillover
# benefit = coef * (max_possible_distance - actual_distance).
# But without a clear "no spillover" reference, use the simpler approach:
# The average control school's absence rate is reduced by spillover.

# Approximate the average spillover effect as the coefficient  $\times$  avg_distance
# relative to some baseline.

# SIMPLEST: The problem asks to use the Q6 estimates.
# Spillover benefit for average control school vs. no-spillover counterfactual:
# If we assume the farthest control school gets ~zero spillover,
# then average spillover = coef * (max_dist - avg_dist)
# Or: use the simple regression coefficient and note that each km closer
# gives |coef| reduction. Average school is avg_dist km away.
# Benefit relative to a school at, say, 10 km: coef * (10 - avg_dist)

# I'll compute it relative to a school at max observed distance (conservative):
spillover_benefit_per_school = abs(coef_dist) * (max_dist_control - avg_dist_control)
student_days_spill_per_school = spillover_benefit_per_school * avg_n_students * 200
total_student_days_spill = student_days_spill_per_school * n_control

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print(f" Max distance among controls: {max_dist_control:.2f} km")
print(f" Spillover benefit per control school")
print(f" (relative to farthest control school): {spillover_benefit_per_school:.4f}")
print(f" Student-days gained per control school: {student_days_spill_per_school:,.1f}")
print(f" Total spillover student-days ({n_control} control schools):
{total_student_days_spill:,.1f}")

# Also compute using a different approach: average reduction relative to "no treatment"
# Assuming no spillover at distance = max
# Total spillover = sum over control schools of coef * (max_dist - school_dist)
total_spill_v2 = 0
for _, row in control_df.iterrows():
benefit = abs(coef_dist) * (max_dist_control - row['distance_nearest_treat'])
total_spill_v2 += benefit * row['num_students'] * 200
print(f"\n Alternative (using each school's actual students and distance):")

print(f" Total spillover student-days: {total_spill_v2:,.1f}")

# (d) Cost-effectiveness
print(f"\n(d) Cost-Effectiveness:")
# Direct only
cost_per_sd_direct = total_cost / total_student_days_treat
print(f" (i) Direct effects only:")
print(f" Total cost: ${total_cost:,.2f}")
print(f" Total student-days gained: {total_student_days_treat:,.1f}")
print(f" Cost per student-day: ${cost_per_sd_direct:.2f}")

# Including spillovers
total_student_days_all = total_student_days_treat + total_student_days_spill
cost_per_sd_all = total_cost / total_student_days_all
print(f"\n (ii) Including spillover effects:")

print(f" Total student-days gained: {total_student_days_all:,.1f}")
print(f" Cost per student-day: ${cost_per_sd_all:.2f}")
print(f"\n Improvement: Cost per student-day decreases by {((cost_per_sd_direct -
cost_per_sd_all)/cost_per_sd_direct)*100:.1f}%")
print(f" when accounting for spillovers.")

print("\n" + "="*80)
print("ANALYSIS COMPLETE")
print("="*80)

```